



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech..

Mid Semester Examinations, October-2024

Course Name: Computer Programming and Problem Solving

Course Code: CS100

Date: 10/10/2024

Time: 10.00 A.M.-11.30 A.M.

Duration: 1.5 Hours

Max. Marks: 50

ANSWER ALL QUESTIONS

- Q.1 a** Evaluate the following expressions: [04 Marks]
- i) $22 + 3 < 6 \ \&\& \ !5 \ || \ 22 == 7 \ \&\& \ 22 - 2 > +5$
- ii) ii) $a + 2 > b \ || \ !c \ \&\& \ a == d * a - 2 < = e$ Where, $a=11, b=6, c=0, d = 7$ and $e=5$.
- b** Convert the following mathematical expression into C equivalent [04 Marks]
- i) $area = s(s-a)(s-b)(s-c)$
- ii) $x = -b + b^2 - 4ac$
- c** Write a program that accepts a number from 4 to 9 only and prints its square using a functional call by value. [06 Marks]
- d** Write a program to find the largest of three numbers using ternary operator. [06 Marks]
- Q.2 a** What is an array? What are the merits and demerits of array in C? [03 Marks]
- b** Differentiate a call by value and call by reference with a suitable example. [03 Marks]
- c** What is a nested loop? Write a program to display multiplications tables from 1 to n. Note: n is an input given by a user. [06 Marks]
- d.** Write a program using pointers to read an array of integers and print its elements in reverse order. [06 Marks]
- Q.3 a** Discuss the advantages of the following: [06 Marks]
- 1) Function
- 2) Pointer
- 3) Array
- b.** Write a program to find the GCD of two numbers using recursion. [06 Marks]



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, Goa-403703

Programme Name: B.Tech

Mid Semester Examination, October-2024

Course Name: Physics

Date: 14.10.2024

Duration: 1 Hour 30 Minutes

Course Code: PH100

Time: 2.00-3.30 pm

Max. Marks: 50

Roll no	2	4	C	5	E	10	2	7
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ANSWER ALL QUESTIONS

- What are the implications of ψ and $|\psi|^2$? Obtain Schrodinger's Time-Independent wave equation and Time-Dependent equation (8M)
- Explain the following failures of Classical Mechanics and how does Quantum Mechanics overcome these failures: (2M+4M+2M)
 - Stability of an Hydrogen Atom
 - Black Body Radiation
 - Photoelectric Effect
- Prove that the following wavefunctions are satisfying the wave equation (6M)

(i) $\psi = \psi_0 \text{Exp} (i (wt - kx))$ and (ii) $Y = A e^{-i\omega(t - \frac{x}{v})}$
- A hydrogen atom is $5.3 \times 10^{-11} \text{m}$ in radius. Use the uncertainty principle to estimate the minimum energy an electron can have in this atom. (5M)
- What de-Broglie wavelength is associated with an electron accelerated through a potential difference of 10000 volts? (Electron mass $m = 9.1 \times 10^{-31} \text{kg}$, Planks constant $h = 6.62 \times 10^{-34} \text{Js}$) (4M)
- An atom in an excited state temporarily stores energy. The lifetime of this excited state is measured to be $1.0 \times 10^{-10} \text{s}$. What is the minimum uncertainty in the energy of the state in eV? (5M)
- Derive the Compton formula. In a Compton collision with an electron, a photon of violet light ($\lambda = 400 \text{ nm}$) is backward scattered through an angle 180° . (8M)
 - How much energy is transferred to the electron in this collision?
 - Compare the result with the energy the electron would acquire in a photoelectric process with the same photon.
 - Could violet light eject electrons from a metal by Compton collision? Explain.
- (a) Show that at low temperatures the Planks formula for radiated energy $E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \times \frac{1}{e^{hc/\lambda kT} - 1} d\lambda$ reduces to the Wein's law i.e. $E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \times e^{-hc/\lambda kT} d\lambda$
 (b) Show that at large temperatures the Planks formula for radiated energy $E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \times \frac{1}{e^{hc/\lambda kT} - 1} d\lambda$ reduces to the Rayleigh-Jeans approximation i.e. $E_\lambda d\lambda = \frac{8\pi kT}{\lambda^4} d\lambda$ (6M)

All the best



Roll No	2	4	C	S	E	1	0	2	7
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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

Mid Semester Examinations, Oct-2024

Section: B

Course Name: Liberal Arts

Date: 10/10/2024

Duration: 1.5 Hours

Course Code: HU100

Time: 2.00 pm- 3.30 pm

Max. Marks: 50

ANSWER ALL QUESTIONS

Answer in One Word

10*1=10

1. In which year was the play *Tughlaq* first published?
2. What language was *Tughlaq* originally written in?
3. Which city does Tughlaq move his capital to?
4. Why did Muhammad bin Tughlaq introduce copper currency?
5. Who poses as a religious leader and opposes Tughlaq in the play?
6. Who is traditionally considered the author of the *Natyashastra*?
7. Approximately how many chapters does the *Natyashastra* contain?
8. What is the central concept of aesthetic experience discussed in the *Natyashastra*?
9. The *Natyashastra* divides drama into how many types?
10. Which rasa represents heroism and energy in the *Natyashastra*?

Match the following

6*1=6

11.

Column A (Characters)	Column B (Descriptions/Actions)
Muhammad bin Tughlaq	Acts as a religious leader who is used as a political pawn
Aziz	Tughlaq's childhood friend who rebels against him
Sheikh Imam-ud-din	Historian who critiques Tughlaq's rule
Ain-ul-Mulk	A cunning opportunist who exploits Tughlaq's policies
Barani	Sultan's confidant and prime minister
Najib	Moves the capital from Delhi

Fill in the Blanks

5*1=5

12. The *rasa* that represents love or beauty is called _____.
13. The word *Natya* in *Natyashastra* refers to _____.
14. _____ is the emotion of peace and tranquility mentioned in the *Natyashastra*.
15. The historical period depicted in the play *Tughlaq* is set in the _____ century.
16. The play portrays the downfall of Tughlaq due to his inability to balance _____ with practical governance.

Answer the following questions in detail:

17. Discuss the theme of idealism versus realism in “Tughlaq.” How does Tughlaq’s vision for his empire clash with the harsh realities he faces? (06)
18. Explain the Rasa theory as outlined in the *Natyashastra*. How do the Rasas enhance the emotional experience of a performance? (07)
19. Identify and discuss the symbolism used in “Tughlaq.” How do specific symbols enhance the themes and messages of the play? (06)
20. Describe the guidelines for stagecraft as provided in the *Natyashastra*. How does the text outline the construction of the stage? (04)
21. Examine the role of secondary characters in “Tughlaq,” such as Tughlaq’s advisors, the common people, and his stepmother. How do these characters contribute to the development of the main themes in the play? (06)



Roll no	2	4	C	S	E	1	0	2	7
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National Institute of Technology Goa

Cuncolim, South Goa - 403 703

Name: B. Tech -I Semester

Mid Semester Examinations –Oct 2024

Course Name: Engineering Mechanics

Date: 14/10/2024

Duration: 90 min

Course Code: ME100

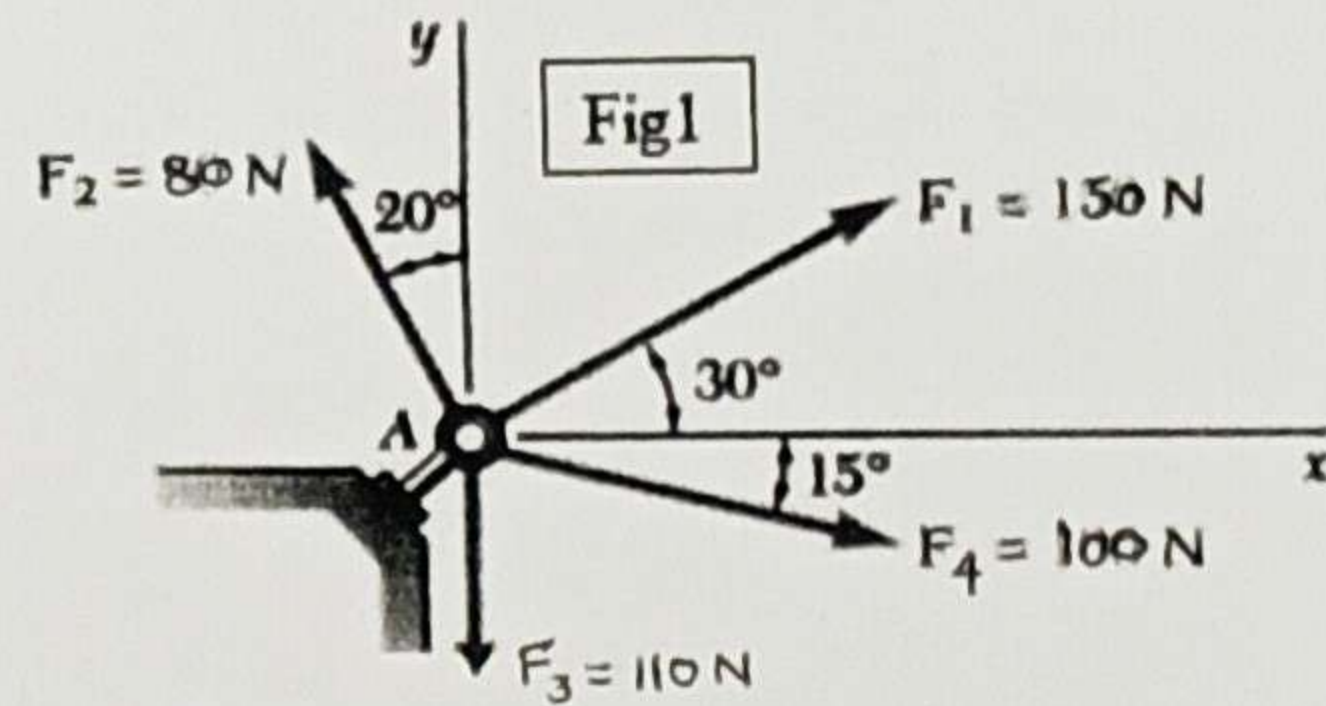
Time: 9.30 am-11 am

Max. Marks: 50

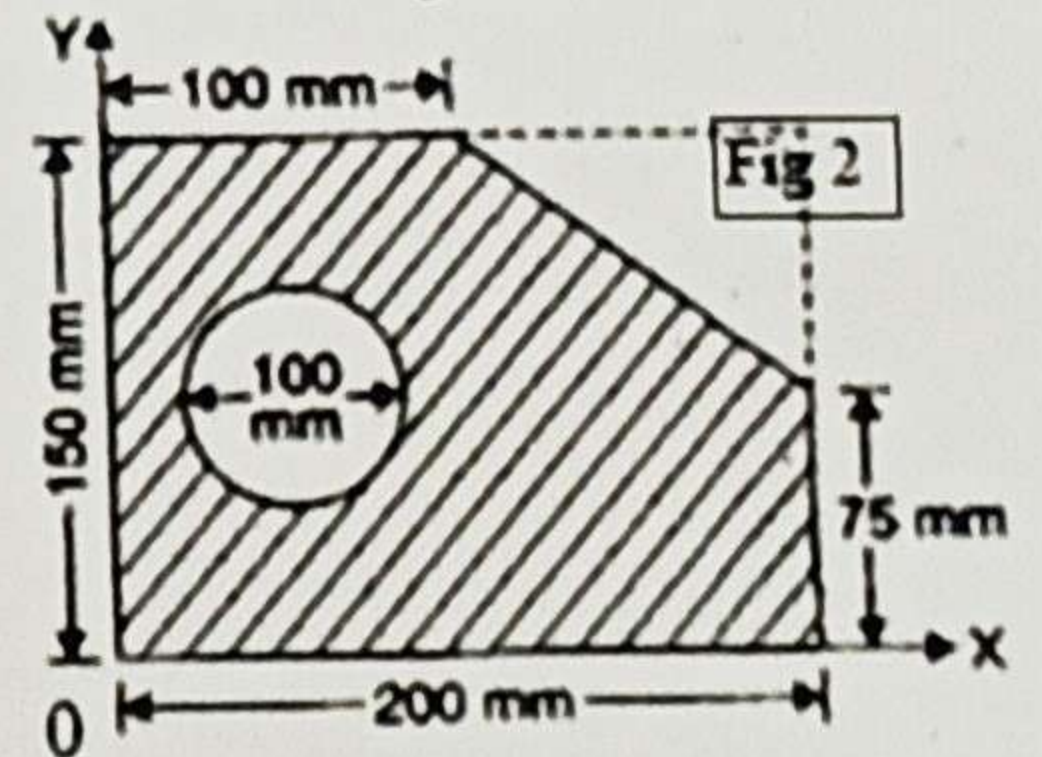
ANSWER ALL QUESTIONS

1. a) Explain the concept of idealization of body with an example.
1. b) Find the magnitude of the two forces, such that if they act at right angles, their resultant is $\sqrt{10}$ N.
But if they Act at 60° , their resultant is $\sqrt{13}$ N (3+3=6M)

2. Forces F_1 , F_2 , F_3 , and F_4 are acting at point A as shown in Fig 1.
1. Determine magnitude and direction of their resultant. (6M)

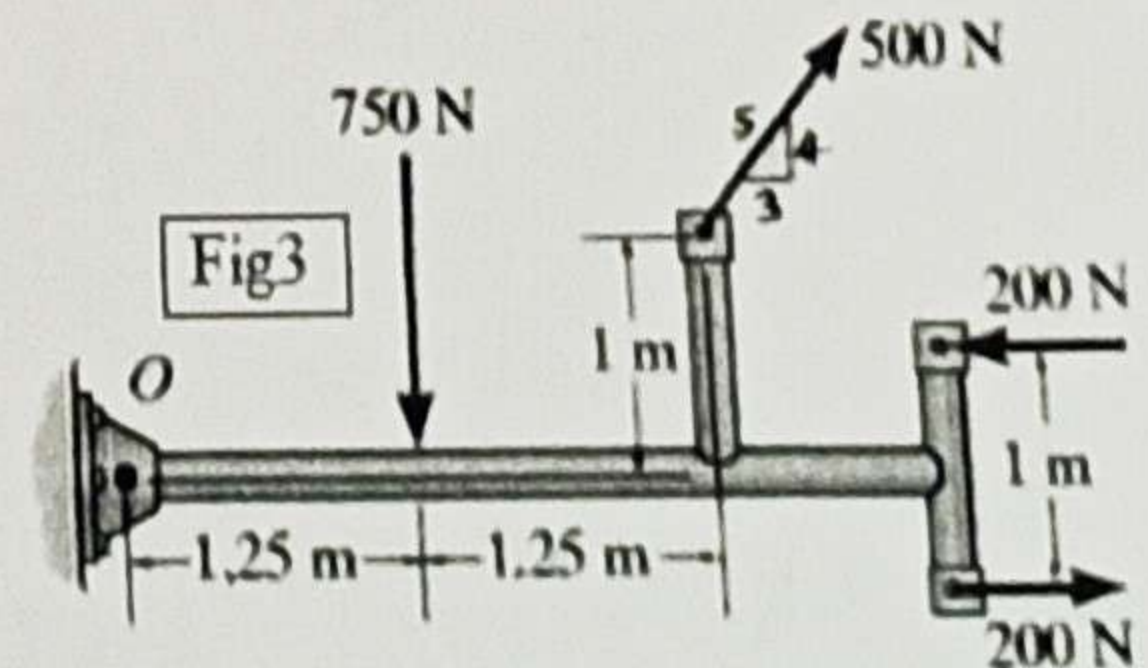


3. a) Find the coordinates of Center of a 100mm diameter hole cut in a thin plate so that this point will be the centroid of the remaining shaded area shown in the Fig 2.

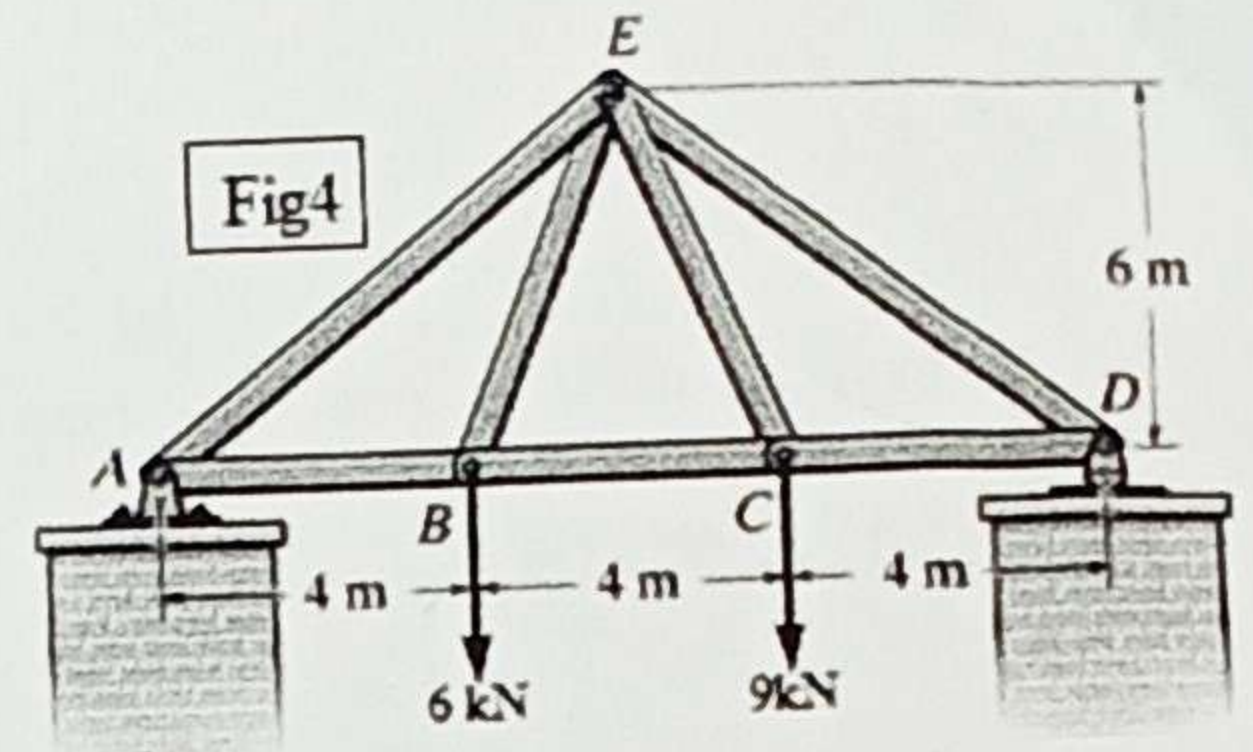


3. b) State Parallel axis theorem (6+2= 8M)

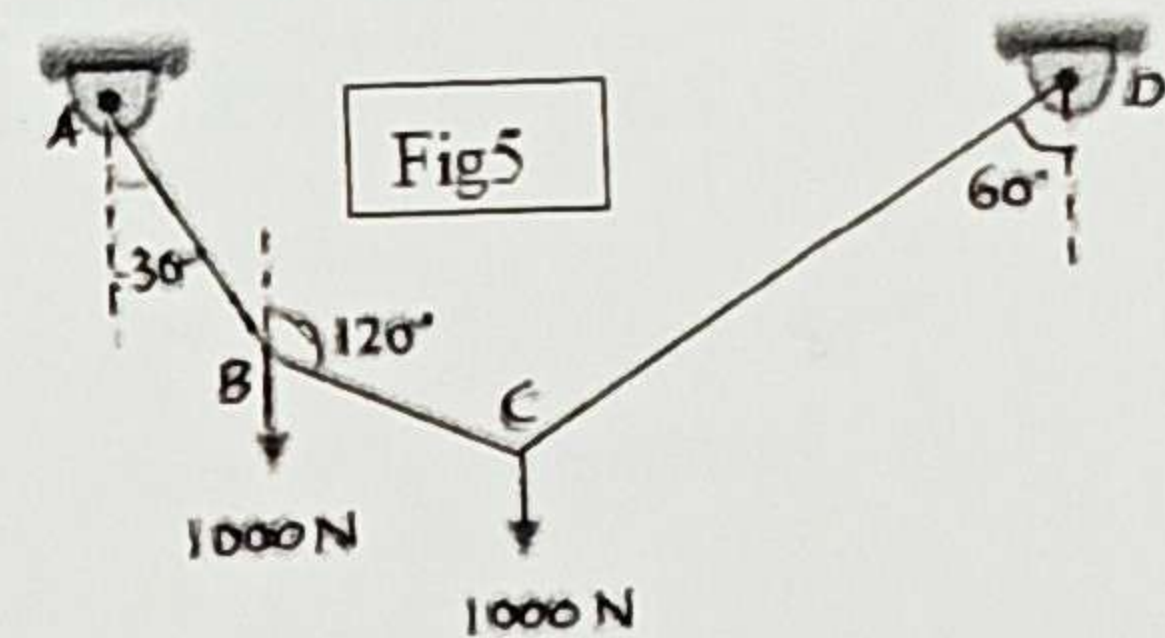
4. a) State Varignon's theorem.
4. b) Determine the resultant moment with respect to point O as shown in Fig 3. (2+4= 6M)



5. Determine the force in member AB, CD and BC of the truss shown in Fig 4 . Also state the nature of the member. (9M)

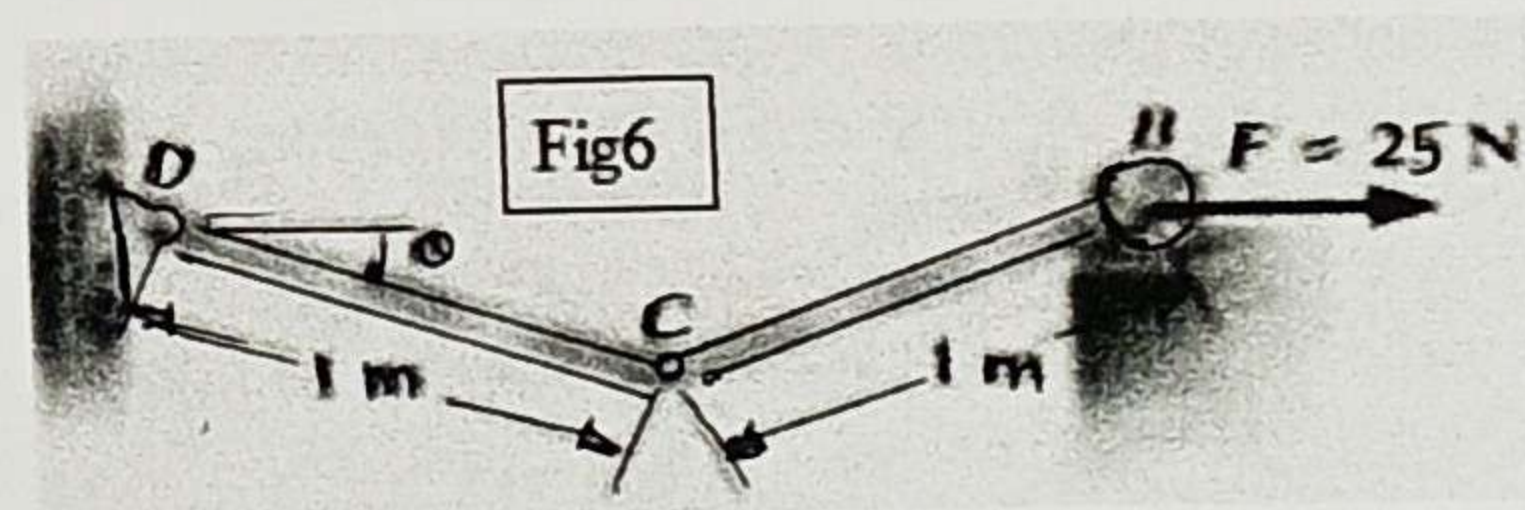


6. A string ABCD, attached to fixed points A and D has two equal weights of 1000 N attached to it at B and C. The weights rest with the portions AB and CD inclined at angles as shown in Fig 5. Find the tensions in the portions AB, BC and CD of the string, if the inclination of the portion BC with the vertical is 120° . (6M)



7. A load of 1.5 kN, resting on an inclined rough plane, can be moved up the plane by a force of 2 kN applied horizontally by a force 1.25 kN applied parallel to the plane. Find the inclination of the plane and the coefficient of friction. (4M)

8. Determine the angle θ for equilibrium of the two member linkage shown in Fig 6. Mass of each member is 10kg. (5M)



-----End-----



Roll No

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राष्ट्रीय प्रौद्योगिकी संस्थान गोवा
NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Goa, 403703

Programme Name: B.Tech.

Mid Semester Examinations, October-2024

Course Name: Basics of Electrical Engineering

Date: 11.10.2024

Duration: 1.5 hour

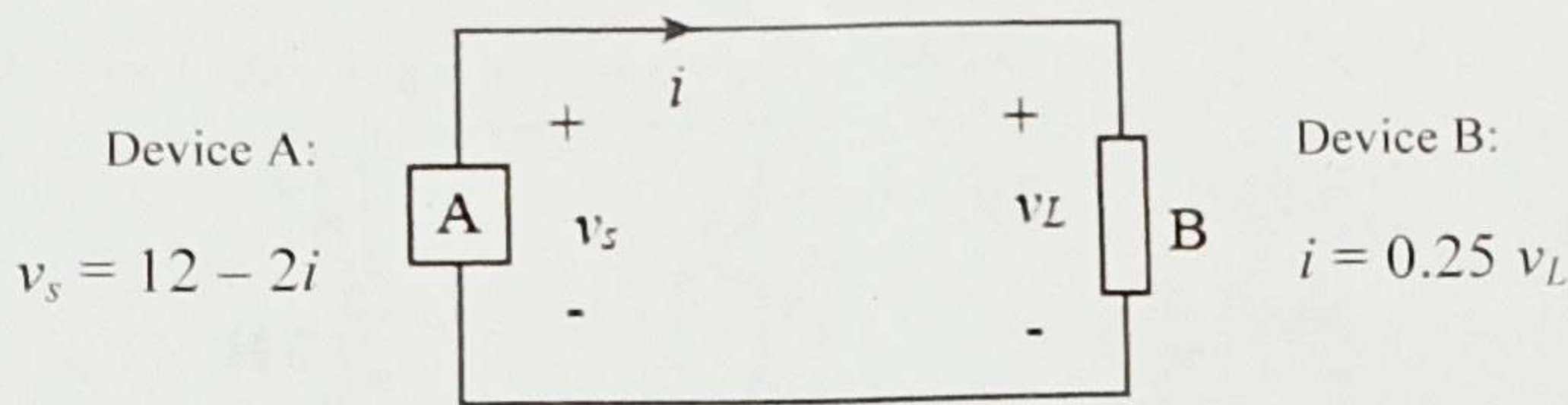
Course Code: EE100

Time: 2:00 pm-3:30 pm

Max. Marks: 50

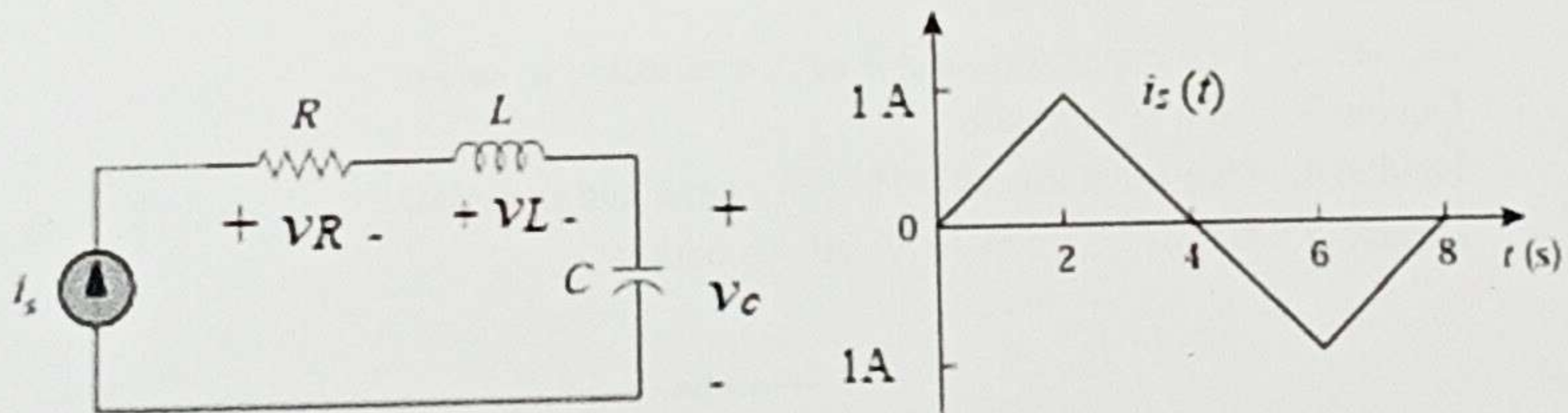
ANSWER ALL QUESTIONS

1. Device A and B are connected as shown in Figure. The voltage current characteristics of these devices are also shown in Figure.



- (a) Using an ideal voltage source and an ideal resistor, built a device whose terminal characteristics is same as that of device A
- (b) Using an ideal current source and an ideal resistor, built a device whose terminal characteristics is same as that of device A
- (c) Determine i and the power dissipated in device B [3x2=6]

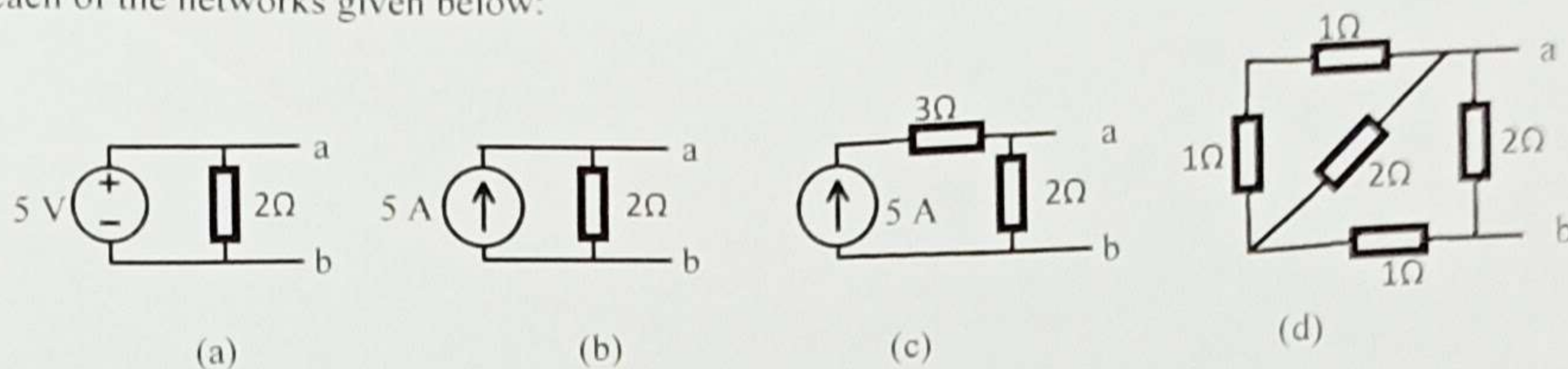
2. The current flowing through the series RLC circuit (shown in Figure) for $0 < t < 8s$ is given in the Figure below. Given that $v_c(0^-) = 1V$. Also given that $R = 5\Omega$, $L = 2H$, and $C = 3F$.



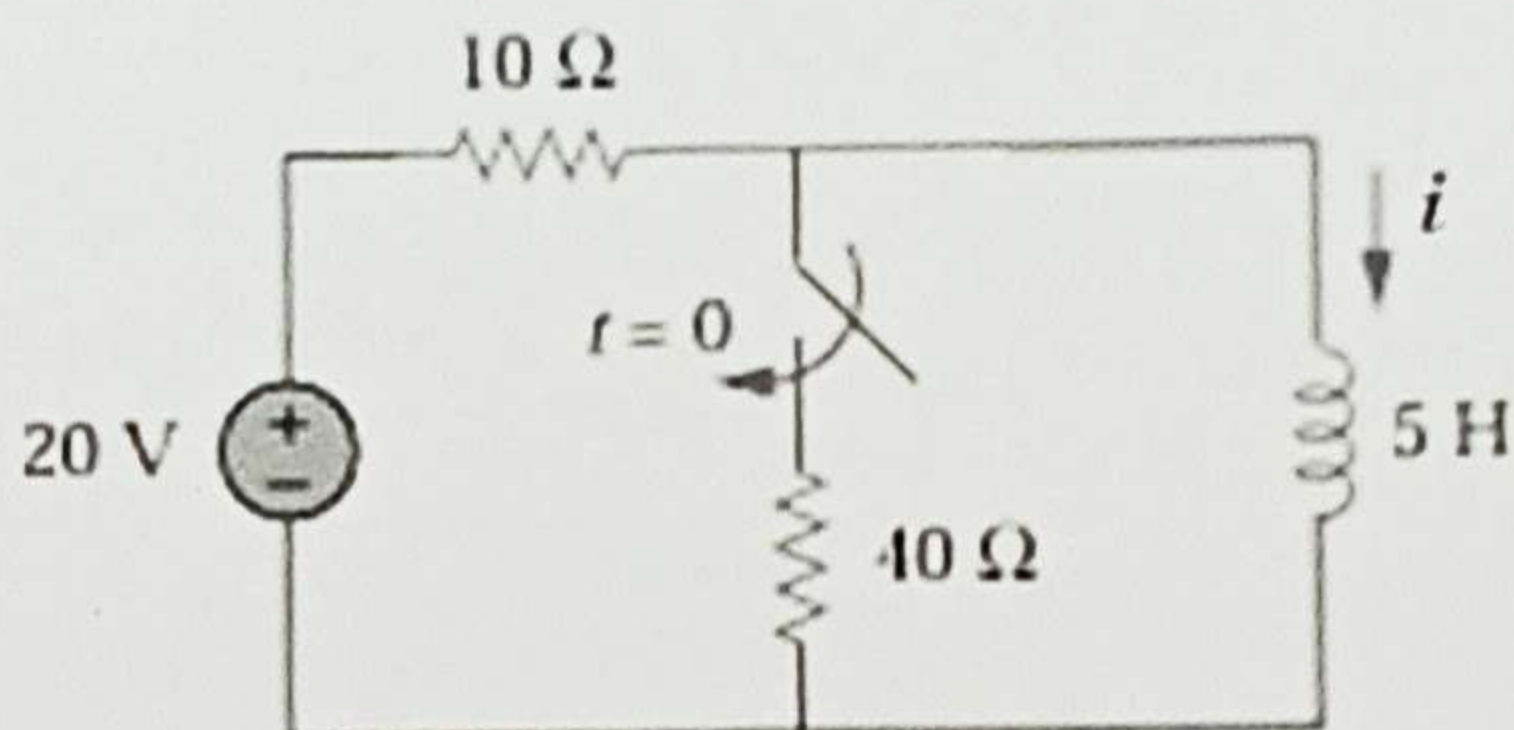
- (a) Determine the value of v_L at $t = 4s$ and $7s$
- (b) Determine the value of v_R at $t = 4s$ and $8s$
- (c) Determine the value of v_C at $t = 4s$ and $8s$
- (d) Determine the maximum amount of energy stored in the inductor and capacitor during the period (0-8 s) [4x2=8]

3. Give examples of any two linear circuit elements with their voltage current characteristics. [3]

4. Find the Thevenin equivalent voltage and resistance (V_{Th} and R_{Th}) as seen across terminals a and b for each of the networks given below: [4x2=8]



5. *Given that $i(0) = 1$ A.* The switch shown in Figure was in open position for a very long duration of time. The switch is closed at $t = 0$ s.

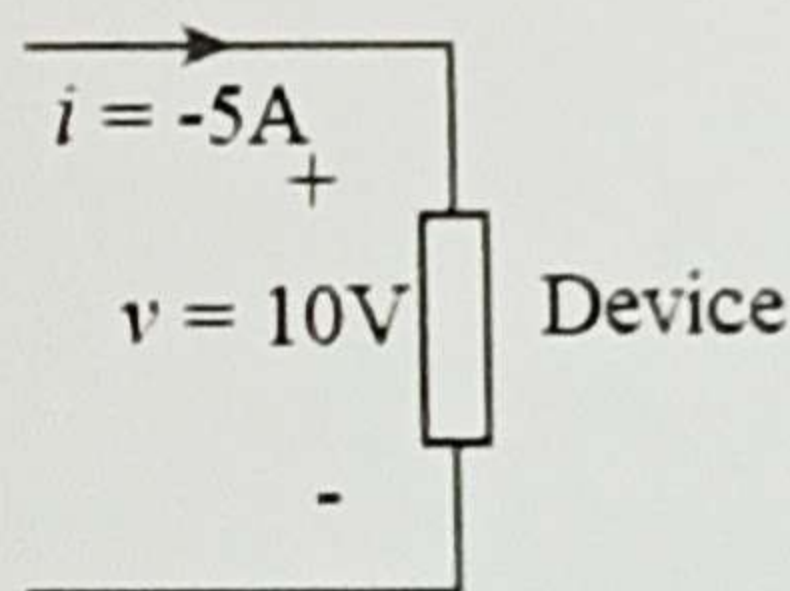


- (a) Determine the time constant of the network which is formed at $t = 0$ s
 (b) Determine ~~$i(0)$~~ , $i(0^+)$, $i(\infty)$, $i(10/8$ s), and $i(0.125$ s)

[7]

6. The voltage across each of the devices described below is equal to 10 V and the current flowing through it is -5 A.

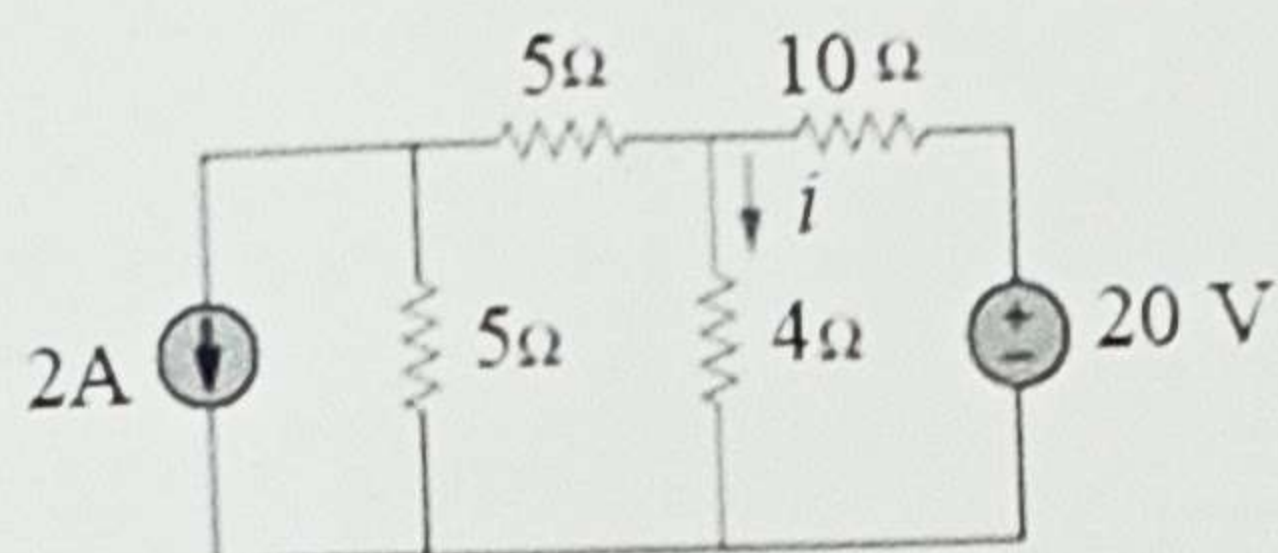
- Device 1: A 2 F capacitor
 Device 2: Two capacitors of 2 F each connected in series.
 Device 3: An 8 H inductor
 Device 4: Two inductors of 8 H each connected in series.
 Device 5: A fully charged 10 V, 10 Ah battery



- (a) Determine the amount of energy stored in each of the above devices.
 (b) Arrange these devices in increasing order of energy stored.

[5+1=6]

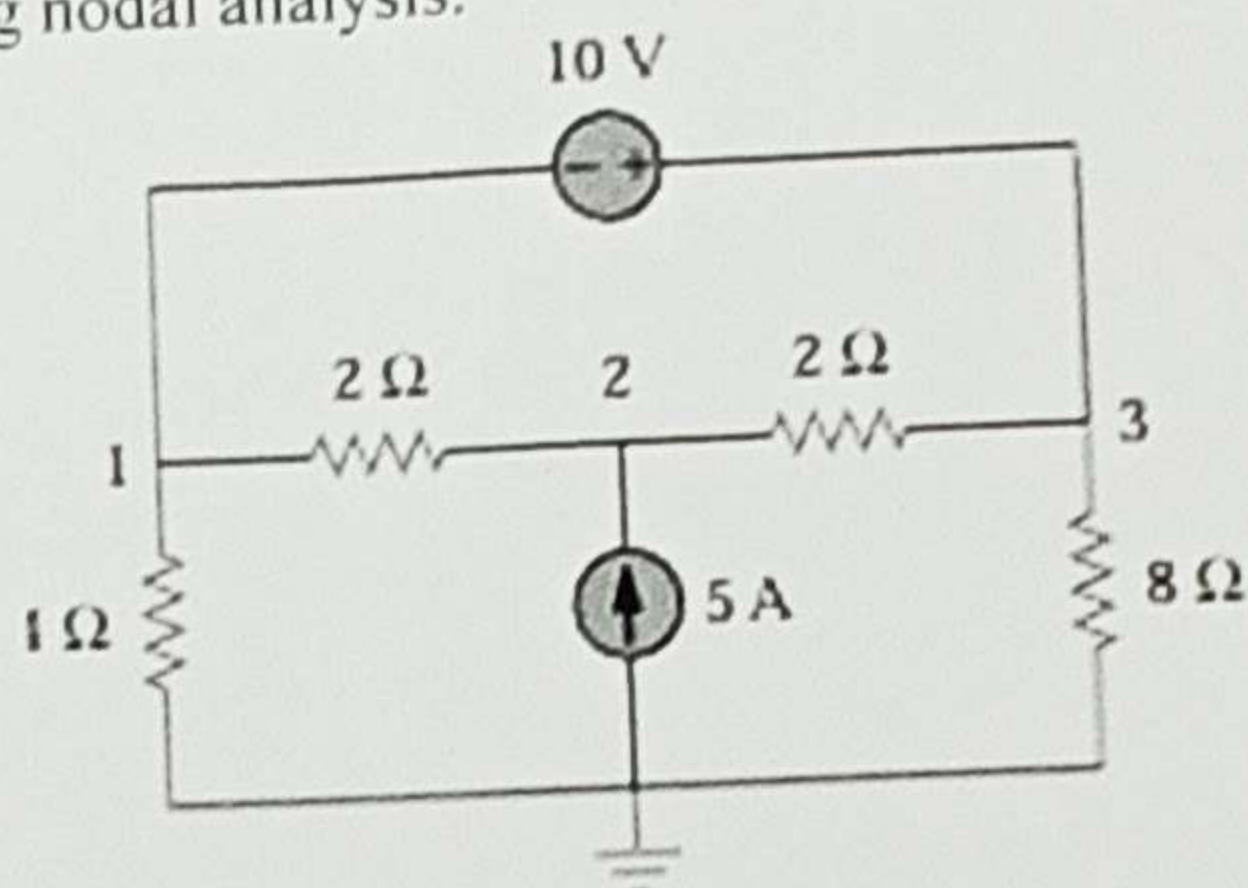
7. For the given circuit:



- (a) Find i using superposition theorem
(b) Verify the result obtained in (a) using series of source transformations

[6]

8. Determine node voltages using nodal analysis:



[6]

-----Best wishes-----



Roll No	2	4	C	5	E	10	2	7
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National Institute of Technology Goa

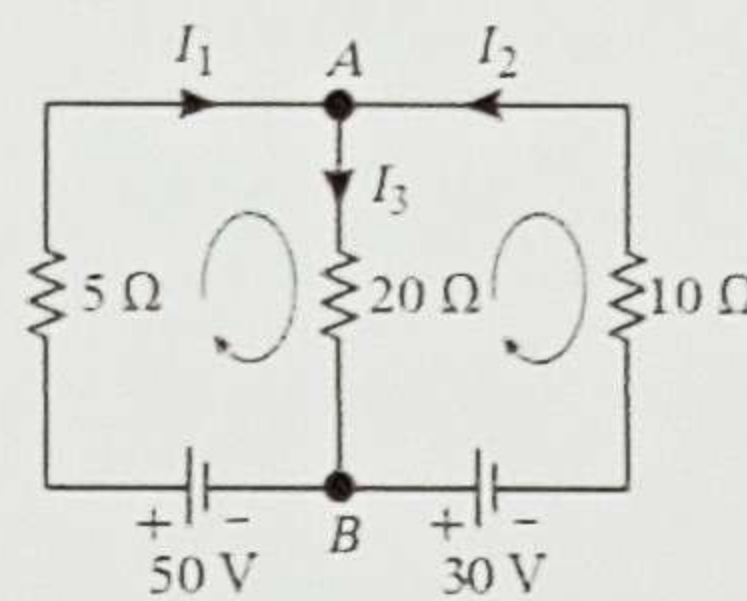
Programme Name: B.Tech. Ist year
Mid Semester Examinations, October 2024

Course Name: **Matrices and Advanced Calculus**
Date: 11/10/2024
Duration: 1 Hour 30 Minutes

Course Code: **MA100**
Time: 10.00 AM - 11.30 AM
Max. Marks: 50

1. Answer All Questions.
2. No marks will be given if the explanation of your answer is missing.
3. The question paper consists of **two** pages.

1. Determine the currents I_1 , I_2 and I_3 in the circuit shown in the following figure using system of equations. [5]



2. Solve the following system of equations using LU-decomposition method [5]

$$3x - 6y - 3z = -3, \quad 2x + 6z = -22, \quad -4x + 7y + 4z = 3.$$

3. Find the inverse of matrix using Gauss-Jordan method [5]

$$P = \begin{bmatrix} 4 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

4. Determine the values of b such that the rank of A is 3. [5]

$$A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ b & 2 & 2 & 2 \\ 9 & 9 & b & 3 \end{bmatrix}$$

5. Find Eigenvalues and Eigenvectors of following matrix

[5]

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

6. Construct a matrix whose nullspace consists of all combinations of $(2, 2, 1, 0)$ and $(3, 1, 0, 1)$.
[5]

7. Find the complete solution $x = x_p + x_n$ for $AX = B$, where

[5]

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 4 & 8 \\ 4 & 8 & 6 & 8 \end{bmatrix}, B = \begin{bmatrix} 4 \\ 2 \\ 10 \end{bmatrix}$$

8. What linear combination of $(1, 2, -1)$ and $(1, 0, 1)$ is closest to $b = (2, 1, 1)$?

[5]

9. Prove or disprove the following

[10]

- (a) If the columns of a matrix are dependent, so are the rows.
- (b) The column space of $2A$ equals the column space of A .
- (c) If the columns of S (eigenvectors of A) are linearly independent, then A is invertible.
- (d) A and A^T have the same left null space.
- (e) If the only eigenvectors of A are multiples of $(1, 4)$ then A has no inverse.

ALL THE BEST